

CATHOLIC JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1
Preliminary Examinations

MATHEMATICS

8863/01

Paper 1

03 Sept 2009

3 hours

Additional Materials: Answer Paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, arrange your answers in NUMERICAL ORDER.

Place this cover sheet in front and fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Name: _____

Class: _____

Question No.	Marks
1	/ 5
2	/ 7
3	/ 9
4	/ 9
5	/ 10
6	/ 5
7	/ 7
8	/ 7
9	/ 9
10	/ 9
11	/ 9
12	/ 14
Total	/ 100

This question paper consists of **5** printed pages.

[TURN OVER]

Section A: Pure Mathematics [40 marks]

- 1 The variables V and x are related by the equation $V = (x^2 + 1)^5$. Given that both V and x vary with time and the rate of increase of V is 320 units per second, find

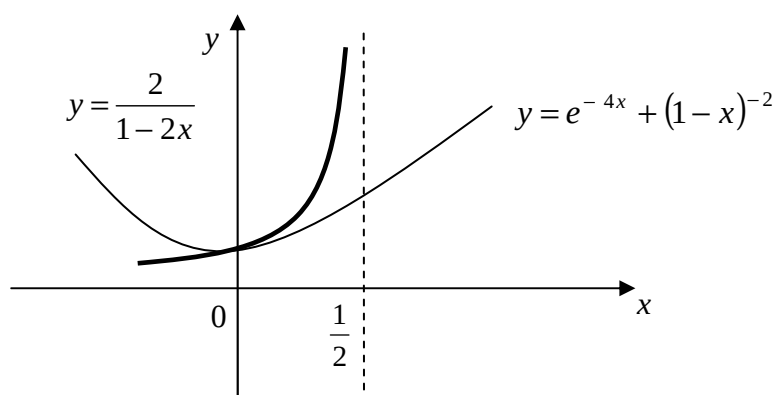
- (i) the rate of increase of x when $x = 1$ unit, [3]
 (ii) x when the rate of increase of x is 100 units per second. [2]

- 2 (i) Find

(a) $\int \frac{1}{1-2x} dx$, [1]

(b) $\int e^{-4x} + (1-x)^{-2} dx$. [2]

- (ii)



The diagram shows part of the graphs of $y = \frac{2}{1-2x}$ and $y = e^{-4x} + (1-x)^{-2}$.

Using your answer in (i), show that the region bounded by the curves $y = \frac{2}{1-2x}$,

$y = e^{-4x} + (1-x)^{-2}$ and the line $x = \frac{1}{4}$ is $\ln 2 + \frac{1}{4e} - \frac{7}{12}$. [4]

- 3 Find, in exact form, the range of values of k for which the curve $y = x^2 - 2kx + 1 - k^2$ meets the x -axis at most once. [3]

- (i) Sketch the graph of $y = \sin\left(\theta - \frac{\pi}{4}\right)$ for $0 \leq \theta \leq 2\pi$, indicating clearly all axial intercepts. [2]

- (ii) Hence, find the range of values of θ for which the curve $y = x^2 - 2x \sin\left(\theta - \frac{\pi}{4}\right) + 1 - \sin^2\left(\theta - \frac{\pi}{4}\right)$ meets the x -axis at most once. [4]

- 4 A curve has the equation $y = x + \ln(x^2 - 4)$, $x \in \mathbb{R}$, $x < -2$ or $x > 2$.
- (i) Using differentiation, find the stationary point on the curve. Determine the nature of this stationary point, explaining clearly how you arrive at your answer. [4]
 - (ii) Find the range of values of x for which the curve is increasing. [2]
 - (iii) Find the equation of the normal to the curve at the point when $x = -4$, leaving your answer in the form $y + ax = \ln b + c$ where a , b and c are integers. [3]
- 5 Functions f and g are defined by
- $$f : x \rightarrow \frac{2}{x-1}, \quad x \geq 0,$$
- $$g : x \rightarrow 3x^2 - 1, \quad x < 0.$$
- (i) Sketch on separate diagrams, the graphs of $y = f(x)$ and $y = g(x)$, indicating clearly all x and y intercepts with the axes and asymptotes (if any). [4]
 - (ii) Explain why g^{-1} exists and express g^{-1} in a similar form. [3]
 - (iii) Explain clearly why $fg(x)$ does not exist. [2]
 - (iv) State the maximal restricted domain for g so that $fg(x)$ exists. [1]

Section B: Statistics [60 marks]

- 6
- (i) Let X be a continuous random variable. State two conditions needed for X to be well modelled by a normal distribution. [2]
 - (ii) Suppose X has the distribution $N(\mu, 25)$. If $P(X < 22) = 0.242$, find the value of μ . [3]
- 7 A particular brand of soft drink is packaged into bottles of 250 ml. A random sample of 80 bottles is taken and the volumes of soft drink in the bottles are measured. The volumes, x ml, are summarized by $\sum(x + 30) = 22560$ and $\sum(x - \bar{x})^2 = 10888$.
- (i) Find unbiased estimates of the population mean and variance. [2]
 - (ii) Assuming a normal distribution, it is concluded that the volume of soft drink in each bottle is understated. State the null and alternative hypothesis clearly, and find an inequality satisfied by the significance level of the test. [4]
 - (iii) State, giving a reason, whether it is necessary to assume a normal distribution for the test to be valid. [1]

- 8** The operator of a section of motorway toll road records its weekly takings according to the types of vehicles using the motorway. For the purpose of charging, there are three types of vehicles: cars, buses and lorries. The weekly takings (in thousands of dollars) for each type are assumed to be normally distributed. These distributions are independent of each other and are summarized in the table.

Vehicle type	Mean	Standard Deviation
Cars	60.2	5.2
Buses	33.9	6.3
Lorries	52.4	4.9

- (i) Find the probability that the weekly takings for lorries are less than the weekly takings for cars. [2]
- (ii) Find the probability that over a 4-week period, the total takings for cars exceeds \$225,000. [2]
- (iii) The operator allocates part of the takings for repairs each week. This is determined for each type of vehicle according to estimates of the long-term damage caused. It is calculated as follows: 5% of the takings for cars, 10% for buses and 20% for lorries. Find the probability that in any given week, the total amount allocated for repairs will not be more than \$20,000. [3]

- 9** The table below shows the results of a survey on 1020 residents.

	Male	Female
Single	170	146
Married	312	314
Widowed	8	42
Divorced/Separated	10	18

Excludes residents who re-married.

One resident is selected at random.

- (i) Find the probability that the resident selected is a
 (a) female and single, [1]
 (b) single given that she is a female, [2]
 (c) male or single. [2]
- (ii) Given further that among the 1020 residents surveyed, 60.76% among the singles are aged 30 to 34, 31.63% among the married residents are aged 30 to 34, none among the widowed residents are aged 30 to 34 and 3.58% among the divorced/separated residents are aged 30 to 34. A resident aged between 30 and 34 is selected at random. What is the probability that the resident is a single? [4]
- 10** In 2005, 13.5% of school-leavers have university education. A random sample of 80 school-leavers is taken. State, giving a reason, the distribution of the number in the sample that has university education. [2]
- (i) Find the probability that more than 9 school leavers have university education. [1]
- (ii) Using a suitable approximation, find the largest value of school-leavers, m such that the probability of having at most m school-leavers with university education in a sample of 80 is not more than 0.825. [6]

[TURN OVER]

- 11 (a)** A club with 400 members is organizing a Family Day. The manager of the club is interested to find out the activity preferences of its members based on age group. The table below shows the number of males and females in each age group.

	Males	Females
Below 21 years old	15	20
21 – 40 years old	80	100
Above 40 years old	100	85

- (i) Describe briefly how a sample of 80 could be chosen using stratified sampling. **[2]**
- (ii) Give a reason why a stratified sample might be preferable in this context. **[1]**
- (iii) The club decides to extend its Family Day invitation to members of the public. Explain briefly whether it would be feasible to continue using stratified sampling if members of the public are to be included in the survey. **[1]**
- (b)** The weekly time spent by residents at community centres is independent with mean 2.7 hours and standard deviation 1.1 hours.
- (i) A random sample of 100 residents is taken. Find the probability that the mean time spent by these 100 residents at community centres is at least 3 hours. **[2]**
- (ii) A different random sample of 20 residents is taken. Find the probability that the total time spent by these 20 residents at community centres does not exceed 45 hours. State an assumption used in this calculation. **[3]**
- 12** A teacher wanted to investigate the relationship between the length of the time spent on revision for H1 Mathematics Mid Year examination and the actual marks obtained at the examination. A set of data was collected from 10 randomly selected students. The time spent, x (in days), and the marks obtained, y , is given below:

x	4	2	7	3	1	5	6	8	10	9
y	50	30	68	43	90	0	66	70	85	75

- (i) Calculate the correlation coefficient for the data. **[1]**
- (ii) Give a sketch of the scatter diagram for the data, as shown on your calculator. **[2]**
- (iii) With reference to your answers in (i) and (ii), explain why it is not advisable to rely solely on the correlation coefficient to draw any conclusion on the linear relationship between x and y . **[2]**
- (iv) By removing two outliers, calculate the correlation coefficient and the regression line of y on x and the regression line x on y for the revised data. **[3]**
- (v) Based on the revised data, use the appropriate regression line to estimate
- (a) the marks obtained by a student who spent 20 days on the revision, **[2]**
- (b) the number of days spent on revision by a student who scored 55 marks. **[2]**
- (vi) Comment briefly on the reliability of the estimates in part (v). **[2]**